Fiscal Year:	FY 2025	**Task Last Updated:**	FY 04/12/2025
PI Name:	T'ien, James S. Ph.D.		
Project Title:	Flame Spread and Extinction Over Solid Materials		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jst2@case.edu	**Fax:**	FY 216 368-3007
PI Organization Type:	UNIVERSITY	**Phone:**	4408295283
Organization Name:	Case Western Reserve University		
PI Address 1:	Mechanical and Aerospace Engineering		
PI Address 2:	620 Glennan Hall, 10900 Euclid Avenue		
PI Web Page:			
City:	Cleveland	**State:**	OH
Zip Code:	44106-1712	**Congressional District:**	11
Comments:			
Project Type:		**Solicitation / Funding Source:**	
Start Date:	03/01/2023	**End Date:**	08/30/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 08/30/2026 per D. Stocker/NASA-GRC and NSSC information (Ed., 9/19/25). NOTE: End date changed to 08/31/2025 per NSSC information (Ed., 10/9/24).		
Key Personnel Changes/Previous PI:	Update: Dr. Chengyao Li has left the university to industry (4/2024). He remains to be a consultant in the project.		
COI Name (Institution):	Li, Chengyao Ph.D. (Case Western Reserve University) Olson, Sandra Ph.D. (NASA Glenn Research Center) Ferkul, Paul (NASA) Johnston, Michael (NASA)		
Grant/Contract No.:	80NSSC23M0035		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The flammability of solid materials is of interest to NASA as the first line of defense in spacecraft fire safety efforts. In micro- or partial-gravity, the reduced buoyant flows, coupled with low-speed air circulation currents, create an environment quite different than normal Earth gravity. In addition, there are several oxidizer atmospheres under consideration for crew vehicles and habitats, which expand the scope of the problem. A fundamental understanding of flame growth, spread, or decay over solids as a function of gravity, flow velocity, oxygen percentage, pressure, and sample size/configuration can be an important contribution both to combustion science and to the current NASA space exploration initiatives. The present work is from the Solid Fuel Ignition and Extinction – Growth and Extinction Limits (SoFIE-GEL) project. SoFIE is a NASA sponsored project for solid fuel combustion in microgravity. It consists of five sub-projects. GEL (Growth and Extinction) is the first sub-project recently carried out in ISS. The objectives of the GEL (Growth and Extinction Limits) project in SoFIE (Solid Flammability, Ignition and Extinction) are (1) to experimentally determine the flame growth and the extinction characteristics of a bulk solid (PMMA sphere) as a function of flow velocity, oxygen percentage, pressure and the degree of sample internal heating in microgravity, and (2) to develop a numerical model that can compare microgravity results with normal gravity results. The GEL experiment was carried out aboard ISS from January 13, 2023, to March 7, 2024.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Research impact: Microgravity experiments on the extinction of bulk solid spheres in forced flow were carried out in ISS. A flammability map was obtained. Near-limit flame extinction characteristics were discovered. Details of flame quenching at low stretch (low flow velocity) were determined. The effect of solid sample preheating on extinction were quantified. Earth Benefit: Material flammability in low-speed flow is important to spacecraft fire safety. Turning off flow circulation is the first line of defense to fight fire in spacecraft. Diffusion flame extinction is also of general interest to fire on earth.
Task Progress:	The present work is from the Solid Fuel Ignition and Extinction – Growth and Extinction Limits (SoFIE-GEL) project. SoFIE is a NASA sponsored project for solid fuel combustion in microgravity. It consists of five sub-projects. GEL (Growth and Extinction) is the first sub-project recently carried out in the International Space Station (ISS). The objectives of the GEL (Growth and Extinction Limits) project in SoFIE (Solid Flammability, Ignition and Extinction) are: (1) to experimentally determine the flame growth and the extinction characteristics of a bulk solid (PMMA sphere) as a function of flow velocity, oxygen percentage, pressure, and the degree of sample internal heating in microgravity; and (2) to develop a numerical model that can compare microgravity results with normal gravity results. The GEL experiment was carried out aboard the ISS from January 13, 2023, to March 7, 2024. The testing periods were divided into two periods: (1) from January 23, 2023 to May 5, 2023; and (2) from November 30, 2023 to March 7, 2024. The reason to have two periods is because of the overheating of the test chamber in test #23125 at the end of the first period. It led to the replacement of a component that needed to be resupplied from the ground. In GEL, 4-cm diameter Polymethylmethacrylate (PMMA) spheres have been chosen as the test samples. This sample size is large enough to have a well-defined front stagnation flow region that minimizes the flame curvature effect, but small enough to fit into the test chamber in the Combustion Integrated Rack (CIR) aboard the ISS. For these samples, the surface regression distance is limited so that sample shape is approximately retained after each test. So, the samples can be re-used multiple times (typically three). This minimizes the number of samples needed to be sent to the ISS and reduces the astronauts' effort to change samples. The sample is also large enough that the disturbance by the multiple embedded thermocouples is minimal. In total, seventeen PMMA samples were used and a total of 54 tests were performed in GEL. The cumulative microgravity sample burning time is about 7 hours. The test results include pressure, oxygen percentage, flow velocity, radiometer readings, ignition current, flame videos from three cameras; multiple temperatures inside solid sample and in the chamber have been downloaded in the NASA Glenn ISS Payload Operations Center (GIPOC) in the NASA Glenn Research Center. The solid temperature data have further been post-processed to yield the temperature gradient data at the solid surface and the heat conductive heat loss rate into the solid. For each test case, these transient data were compiled to form composite videos that can be used for visualizing the experiment and to analyze the data. So far, the dynamic behavior of the high-speed blowoff limit and the low-velocity quenching limit has been analyzed. The extinction boundary map for the one-atmospheric cases has been established. The effect of the gas heat loss into solid interior has been quantified. Further analysis of the data may include the effect of pressure, enriched oxygen (NASA exploration atmosphere), flame growth rate, near-limit flame temperatures and a comparison with model.
Bibliography Type:	Description: (Last Updated: 09/19/2025)
Abstracts for Journals and Proceedings	Ferkul PV, Hsu S-Y, Huang J-H, Olson SL, Johnston MC, Li C, T'ien JS. "Burning of a solid sphere in microgravity: formation of an unusual diffusion flame." 14th U.S. National Combustion Meeting (USNCM), Boston, Massachusetts, March 17-19, 2025. Abstracts. 14th U.S. National Combustion Meeting (USNCM), Boston, Massachusetts, March 17-19, 2025. , Mar-2025
Abstracts for Journals and Proceedings	Ferkul PV, Hsu S-Y, Huang J-H, Olson SL, Johnston MC, Li C, T'ien JS. "Burning of a solid sphere in microgravity: formation of an unusual diffusion flame." 14th U.S. National Combustion Meeting (USNCM), Boston, Massachusetts, March 17-19, 2025. Abstracts. 14th U.S. National Combustion Meeting (USNCM), Boston, Massachusetts, March 17-19, 2025. , Mar-2025
Articles in Peer-reviewed Journals	James S, Li C, Ferkul PV, Olson SL, Johnston MC. "Experimental determination of diffusion flame extinction boundaries over a PMMA sphere: The effect of sub-surface solid heating." Combustion and Flame. 2025 Jul 1;277:114244. https://doi.org/10.1016/j.combustflame.2025.114244 , Jul-2025
Articles in Peer-reviewed Journals	Li C, T'ien JS, Ferkul PV, Olson SL, Johnson MC. "Extinction of solid diffusion flame: Details of quenching and blowoff processes." Proceedings of the Combustion Institute. 2024 Jan 1;40(1-4):105373. https://doi.org/10.1016/j.proci.2024.105373 , Jan-2024

Articles in Peer-reviewed Journals	Johnston CE, T'ien JS, Hsu S-Y, Wu CW, Olson SL, Ferkul PV. "Quenching extinction of solid sphere diffusion flames induced by a sudden removal of gravity." Fire Safety Journal. 2024 Jun 1;146:104137. https://doi.org/10.1016/j.firesaf.2024.104137 , Jun-2024